

Polina Dolgova

PhD Student, CISPA Helmholtz Center for Information Security (Saarland University)

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Research Interests

- **Machine unlearning:** formal definitions, data preservation/removal, and data attribution.
- **Continual learning:** methods for retaining useful knowledge under model updates.
- **Stability, efficiency and robustness** of learning algorithms
- **Alignment-related** questions of data influence, controllable updates, and reliable model behavior.
- **Mathematical foundations** of machine learning and theory-driven empirical methods.

Publications and Preprints

Forgetting Has Neighbors: Localized Collateral Forgetting in Machine Unlearning.

Polina Dolgova, Sebastian U. Stich.

arXiv preprint, 2026. [arXiv:2605.31317](#)

Summary: Analyzes unintended side effects of machine unlearning, showing that forgetting errors can propagate locally to neighboring data points.

Contribution: proposed the core idea, developed the theoretical analysis, and conducted experiments.

Sequential Subspace Noise Injection Prevents Accuracy Collapse in Certified Unlearning.

Polina Dolgova, Sebastian U. Stich.

arXiv preprint, 2026. [arXiv:2601.05134](#)

Summary: Studies certified machine unlearning and proposes a strategy to preserve utility while satisfying formal unlearning guarantees.

Contribution: proposed the core idea, developed the theoretical analysis, and conducted experiments.

Education

Saarland University

(CISPA Helmholtz Center for Information Security)

PhD in Computer Science

Saarbrücken, Germany

Supervisor: [Sebastian U. Stich](#)

Jul 2025 – present

The National Research University Higher School of Economics (HSE) 2021 – 2023

M.S. in Applied Mathematics and Computer Science

Moscow, Russia

Focus: theoretical computer science

Thesis: Topological Data Analysis in Medical Applications

Supervisor: [Vsevolod Chernyshev](#)

Yandex School of Data Analysis

Moscow, Russia

M.S.-level program in Machine Learning

2019 – 2022

The National Research University Higher School of Economics (HSE) 2016 – 2020
B.S. in Mathematics (with honors)
Moscow, Russia
Thesis: Clique-Chromatic Numbers for Intersection Graphs
Supervisor: [Andrei M. Raigorodskii](#)

Selected Industry Experience

Insud Pharma — ML Engineer (R&D) 2024–2025

- Led applied ML research projects under severe data scarcity and domain shift constraints.
- Designed a data-efficient ASR adaptation pipeline for specialized medical audio with accented speech and domain-specific terminology, combining synthetic data generation and parameter-efficient fine-tuning.
- Investigated robustness and error modes of OCR and anomaly detection systems for pharmaceutical manufacturing, including weakly supervised and low-data settings.
- Combined literature-driven analysis with custom rule-based and learned components for handwritten chemical structure recognition.

Yandex — ML Engineer (R&D) 2021–2024

- Worked on large-scale text generation with LLMs, including reinforcement learning-based training and parameter-efficient fine-tuning.
- Investigated diversity of generated text as a learning objective, studying how training signals, data composition, and model updates affect output behavior.
- Designed and evaluated efficiency-oriented learning approaches under strict latency and memory constraints, analyzing trade-offs between model size, performance, and training dynamics.
- Conducted exploratory research on voice verification using speaker embeddings, clustering, and pseudo-label fine-tuning.
- Mentored junior engineers and supervised research-oriented internship projects.

Research Experience

Certified Machine Unlearning 2025

- Proposed a sequential subspace noise injection method to prevent accuracy collapse in certified unlearning.
- Developed theoretical analysis and validated the approach empirically.

Graph Sparsification as Distillation in Neural Networks 2024

- Adapted theoretical conditions for graph sparsifier existence to computational settings.
- Investigated connections between graph sparsification, distillation, and neural network compression.

Topological Data Analysis in Medical Applications 2022–2023

- Explored TDA-based representations for medical diagnostic tasks and analyzed feature relevance.

Physically Informed Neural Networks for Medical Imaging 2021–2022

- Studied the impact of incorporating physical constraints into neural models for MRI-based diagnostics.

Clique-Chromatic Numbers for Intersection Graphs 2019–2020

- Proposed and proved new estimates for clique-chromatic numbers in intersection graph families.

Talks, Posters and Summer Schools

- Poster (workshop): *Sequential Subspace Noise Injection Prevents Accuracy Collapse in Certified Unlearning*, [PPML @ EurIPS 2025](#).
- Talk: *Sequential Subspace Noise Injection Prevents Accuracy Collapse in Certified Unlearning*, research visit talk, University of Zurich, 2025.
- [CISPA–ELLIS Summer School on Trustworthy AI – Secure and Safe Foundation Models](#), 2025.

Research Visits

University of Zurich (UZH), Zurich, Switzerland
Hosted by [Anastasia Koloskova](#)

Nov 2025

Selected Awards and Honors

- 1st place, Yandex Ads Hackathon, 2023.
- Winner, Vysshaya Liga (national student olympiad, Russia), 2020.
- Winner and prize-winner in nationally recognized high-school olympiads and academic tournaments in mathematics and economics, including individual and team competitions.

Teaching and Service

Coach in Olympiad Mathematics

2017 – 2024

Languages

Russian (native), English (fluent), Spanish (basic conversational)